

Sleep Arousal Detection with Convolutional Neural Networks (Li, Guan)

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Problem Definition

Sleep disorders, including insomnia, sleep apnea, and hypersomnia, affected an estimated $\frac{1}{3}$ of **U.S. adults**. Excessive sleep arousals (brief interruptions of wakefulness during sleep) contribute to fragmented sleep, cardiovascular dysfunction, obesity, depression, and cognitive impairment. Precision medicine can help **make sleep-arousal detection more accurate** by tailoring results to each individual, combining signals with patient-specific factors, leading to **personalized diagnosis and treatment decisions**.

Scalability Analysis

GATK HaplotypeCaller run for N = 1, 2, 4, 8, 16 on Google Cloud Batch. Per-patient runtime constant at **~1,800s**; **98.7% weak scaling efficiency**. **BaseRecalibrator** and **HaplotypeCaller** dominate (**~79%** of runtime). Auto-scaling with **~2-3 min** startup overhead per stage.

Security and Privacy Analysis

Data used in this project comes from the 2018 PhysioNet Challenge and the Sleep Heart Health Study, both **publicly available and properly de-identified**. Any additional records would be de-identified by removing direct identifiers and access to data and activity would be monitored.

Cost Analysis at Scale (per Patient)

Each participant runs on an e2-standard-2 VM (**~\$0.034 compute + ~\$0.01 storage = ~\$0.044/participant**). Full 16-participant run costs **~\$0.70**. **BaseRecalibrator** and **HaplotypeCaller** account for **~79%** of compute cost.

Data types & Bioinformatics Pipelines and Reproducibility prompts results

Data Types

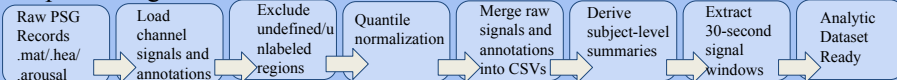
wearables/clinical time series $\rightarrow$ **polysomnography** recordings from PhysioNet
13 channels: (F3-M2, F4-M1, C3-M2, C4-M1, O1-M2, O2-M1, E1-M2, Chin1-Chin2, ABD, CHEST, AIRFLOW, SaO2, ECG)
binary **per-time-point arousal labels**

genomics $\rightarrow$ whole genome sequencing FASTQ
short read sequencing files used for scalability testing

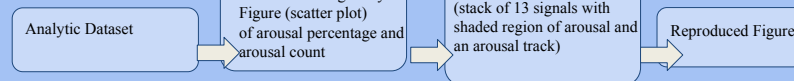
Cloud Service

GCP

Preprocessing DFD



Downstream DFD



Scalability Testing Pipeline

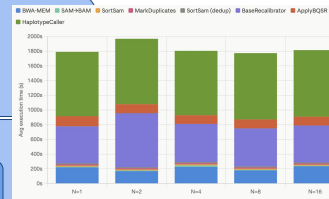


Figure 2b: Arousal Heterogeneity Across Individuals (N=10 subjects)

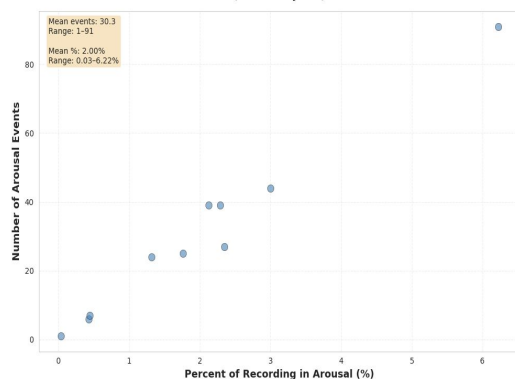


Figure 6b: 30-Second Signal Window with Arousal Events

